

IntelliSpoil: Hybrid Deep Learning for Rapid Detection of Fruits and Vegetables Spoilage Using Autoencoder and CNN

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Abstract

Global food security and economic stability are increasingly threatened by post-harvest losses, with spoilage accounting for billions in annual revenue deficits. This study proposes a cutting-edge deep learning framework that synergizes Convolutional Autoencoders for unsupervised feature extraction with Convolutional Neural Networks (CNNs) for high-precision binary classification of produce quality. By leveraging three massive Kaggle datasets—Fruits 360, Fruit and Vegetable Disease, and Fresh and Stale Classification—this research develops a robust system capable of accurately distinguishing between fresh and rotten fruits and vegetables. The Autoencoder architecture distills complex visual data into meaningful latent representations, while the CNN performs binary classification. This system aims to revolutionize quality control in retail and supply chains, significantly reducing waste, enhancing food safety, and fostering a more sustainable global food ecosystem.

1 Background

Global food waste has reached alarming levels, with approximately one-third of all food produced being lost or wasted annually, leading to severe environmental degradation and economic instability. Fruits and vegetables are particularly vulnerable to spoilage due to their high moisture content, perishable nature, and extreme sensitivity to environmental fluctuations. Traditional quality assessment methods rely on manual inspection, which is inherently time-consuming, subjective, and prone to human error. Recent breakthroughs in computer vision and deep learning have demonstrated transformative potential for automated food quality assessment. Multiple comprehensive datasets including the Fruits 360 dataset [1], Fruit and Vegetable Disease dataset [2], and Fresh and Stale Classification dataset [3] provide rich image collections suitable for training robust machine learning models. Studies have shown that AI-driven systems can effectively detect early spoilage indicators in produce, offering a scalable solution to a global crisis [5, 6].

2 Objectives

1. Engineer a state-of-the-art automated fruits and vegetables spoilage prediction system using advanced deep learning architectures.
2. Implement a high-performance Autoencoder archi-

ture for sophisticated unsupervised feature extraction from diverse fruits and vegetables images.

3. Design a robust CNN classifier to accurately distinguish between fresh and rotten produce.
4. Rigorously evaluate model performance using precision-recall curves, F1-scores, and confusion matrices to ensure industrial-grade reliability.
5. Architect a scalable deployment framework tailored for real-world integration in high-volume retail and global supply chain environments.

3 Data Sources

3.1 Primary Datasets

Three complementary datasets from Kaggle will be utilized to ensure comprehensive training coverage:

Fruits 360 Dataset [1]: Contains over 90,000 images of 131 different fruits and vegetables, captured from multiple angles with consistent lighting and background. Images are available in 100x100 pixel resolution, providing standardized input for baseline model training and feature extraction.

Fruit and Vegetable Disease Dataset [2]: Provides labeled images categorized as healthy versus rotten, covering various fruits and vegetables. This binary classification dataset is essential for training the model to distinguish between fresh and spoiled fruits and vegetables with clear quality degradation patterns.

Fresh and Stale Classification Dataset [3]: Contains images specifically labeled for freshness classification, including fresh and stale categories across multiple produce types. This dataset complements the others by providing a wide variety of examples for robust binary quality assessment.

3.2 Data Augmentation

To simulate various spoilage conditions and improve model generalization, the following augmentation techniques will be applied:

- Rotation and flipping to capture different viewing angles.
- Color jittering to represent varying lighting conditions.
- Gaussian noise addition to simulate camera variations.
- Synthetic spoilage pattern generation using image processing techniques.

3.3 Preprocessing

Data preparation includes:

- Image resizing and normalization to standard dimensions.
- Color space conversion for optimal feature extraction.
- Train-validation-test split (70-15-15 ratio).
- Data balancing techniques to handle class imbalance between fresh and spoiled categories.

4 Methodology

4.1 Autoencoder for Feature Extraction

A convolutional Autoencoder will be trained to learn compressed representations of fruits and vegetables images in an unsupervised manner. The architecture consists of:

- **Encoder:** Multiple convolutional layers with decreasing filter sizes to compress spatial information into a latent vector representation.
- **Latent Space:** Low-dimensional bottleneck capturing essential fruits and vegetables characteristics.
- **Decoder:** Transposed convolutional layers reconstructing the original image from the latent representation.

The Autoencoder learns to capture intrinsic features such as texture, color patterns, and structural integrity that are indicative of produce quality and spoilage across a wide variety of fruits and vegetables.

4.2 CNN Classification Architecture

The trained Autoencoder's encoder will be integrated with a CNN classifier for supervised spoilage prediction. The CNN architecture includes:

- Pre-trained encoder layers (frozen or fine-tuned).
- Additional convolutional layers for hierarchical feature learning.
- Batch normalization and dropout layers for regularization.
- Dense layers with softmax activation for binary classification.

The system will classify fruits and vegetables into two primary categories: Fresh and Rotten, enabling proactive quality management and dynamic pricing strategies.

4.3 Training Strategy

- **Phase 1:** Train Autoencoder using reconstruction loss (Mean Squared Error).
- **Phase 2:** Fine-tune combined architecture using categorical cross-entropy loss.
- **Optimization:** Adam optimizer with learning rate scheduling.
- **Regularization:** Early stopping and model checkpointing based on validation performance.
- **Evaluation:** Cross-validation to ensure model robustness.

5 Expected Outcomes

- A high-fidelity deep learning model achieving higher accuracy in binary spoilage classification (Fresh vs. Rotten).
- Advanced visualization of learned features through Autoencoder latent space analysis, providing insights into the biological markers of decay.
- Enhanced model interpretability through feature importance maps using gradient-based visualization techniques (Grad-CAM).
- A production-ready prototype system with real-time, low-latency prediction capabilities.
- Strategic recommendations for large-scale implementation in automated retail quality control systems.

6 Risks and Mitigation

- **Limited spoiled fruits and vegetables samples:** Addressed through sophisticated data augmentation and transfer learning from large-scale pre-trained models.
- **Computational requirements:** Mitigated by using cloud computing platforms and model optimization techniques.
- **Overfitting:** Controlled through dropout, data augmentation, and regularization techniques.
- **Real-world variability:** Tested on diverse lighting conditions and backgrounds to ensure generalization.
- **Class imbalance:** Handled using weighted loss functions and synthetic minority oversampling.

7 References

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